Class #12 - Orthogonal Projections

1. An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

   Proof:

2. Let U be an $m \times n$ matrix with orthonormal columns, and let x and y be in $\mathbf{R^n}$. Then,

   (a) $||Ux|| =$ _____

   (b) $(Ux) \cdot (Uy) =$ _____

   (c) $(Ux) \cdot (Uy) = 0$ if and only if _____

   Proof:

3. An *orthogonal matrix* is a square _____matrix U such that $U^{-1} =$ _____.
   Fill in the blank: An orthogonal matrix has _____columns.
   Fill in the blank: Any square matrix with orthonormal columns is an orthogonal matrix, and such a matrix has _____rows!

4. Example: Let $\{u_1, u_2, \cdots, u_5\}$ be an orthogonal basis for $\mathbf{R^5}$ and let

$$y = c_1u_1 + c_2u_2 + \cdots + c_5u_5.$$

   Consider the subspace $W = \text{Span}\{u_1, u_2\}$, and write y as the sum of a vector z_1 in W and a vector z_2 in $W^\perp$.

5. The Orthogonal Decomposition Theorem: Let W be a subspace of $\mathbf{R^n}$. Then each y in $\mathbf{R^n}$ can be written uniquely in the form

$$y = \hat{y} + z$$

   where _____is in _____and _____is in _____. In fact, if $\{u_1, \cdots, u_p\}$ is any orthogonal basis of W, then

$$\hat{y} = _____$$

   and $z = y - \hat{y}$.

6. Fill in the blank: If y is in $W = \text{Span}\{u_1, \cdots, u_p\}$, then orthogonal projection of y onto $W =$ _____.

7. The Best Approximation Theorem: Let W be a subspace of $\mathbf{R^n}$, y any vector in $\mathbf{R^n}$, and $\hat{y}$ the orthogonal projection of y onto W. Then _____is the closest point in W to y, in the sense that

$$||y - \hat{y}|| < ||y - v||$$

for all v in W distinct from _____.

Proof:

8. Example: If $u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, $y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, and $W = \text{Span } \{u_1, u_2\}$, then the closest point in W to y is:

9. The distance from a point y in $\mathbf{R^n}$ to a subspace W is defined as the distance from y to the nearest point in W. Find the distance from y to W above.